

1. **SLE vs SNLE.** In some cases, the inverse of a nonsingular matrix needs to be explicitly computed. Give possible reasons why. Finding the inverse of a matrix is equivalent to simultaneous solution of systems of linear equations. Expound why? On another hand, it can be posed as a simultaneous solution of systems of nonlinear equations as presented in Heath (First Ed.) Chapter 5 Computer Problem #5.17. Discuss the derivation of the iterative solution using Newton's method.

Let  $n$  be a random positive integer on  $[5,20]$ . Generate a random  $n \times n$  matrix with integer entries on  $[-10,10]$ . Compare the execution time of your machine for computing its inverse using the following methods:

- 1.1. Gauss-Jordan Reduction
- 1.2. LU Factorization
- 1.3. Iterative Scheme discussed in Computer Problem #5.17

Are the results according to what you expected? Why?

2. **LLSE vs NLLSE.** Kepler's Third Law states that planets move in elliptical orbit around the sun at the center. Given the observations on some planet's position, determine the equation of the planet's orbit then plot its orbit.

|   |      |      |      |      |      |      |      |      |      |      |
|---|------|------|------|------|------|------|------|------|------|------|
| x | 1.02 | 0.95 | 0.87 | 0.77 | 0.67 | 0.56 | 0.44 | 0.30 | 0.16 | 0.01 |
| Y | 0.39 | 0.32 | 0.27 | 0.22 | 0.18 | 0.15 | 0.13 | 0.12 | 0.13 | 0.15 |

- 2.1. Present the problem as a linear curve fitting problem and solve using method of orthogonal transformation. Hint: Use general form of an ellipse.
- 2.2. Present the problem as a nonlinear curve fitting problem and solve using Gauss-Newton method. Hint: Use standard form of an ellipse.

Perturb the data above by adding a random number on  $[-0.005, 0.005]$  in each component of the data points. Redo 2.1 and 2.2 using three different perturbations and plot the orbit of the solutions together with the original problem. What can you say about the curve fitting problem?

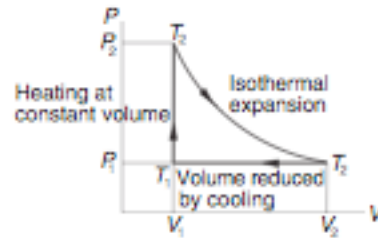
3. **INTERPOLATION.** The specific heat of  $c_p$  in J/gK of a conducting material is dependent on temperature in K. For aluminum, the sepecific heat varies as follows:

|       |        |       |       |       |       |      |
|-------|--------|-------|-------|-------|-------|------|
| T     | -250   | -200  | -100  | 0     | 100   | 300  |
| $c_p$ | 0.0163 | 0.318 | 0.699 | 0.870 | 0.941 | 1.04 |

- 3.1. Perform Newton Interpolation and Natural Cubic Spline Interpolation for the above data and superimpose the interpolations at intervals of 10 K.
- 3.2. Compare the interpolation at  $T=273$  K. Search the internet for the specific heat of aluminum at room temperature ( $0^\circ\text{C}$ ) and compute for the errors of approximation.
- 3.3. Inverse-interpolation can be posed as a root-finding problem. Knowing the interpolant, one can perform root-finding method to identify the x-coordinate given the y-coordinate. Using Regula-Falsi method, identify at what temperature does aluminum reach a specific heat of 1.0. Compare the solutions from both interpolants.

4. **ROOTS.** Present the following problems in Physics/engineering as root-finding problems and solve each using the indicated method of solution.

#### 4.1. Newton Method



The figure shows the thermodynamic cycle of an engine. The efficiency of this engine for monatomic gas is

$$\eta = \frac{\ln(T_2/T_1) - (1 - T_1/T_2)}{\ln(T_2/T_1) + (1 - T_1/T_2)/(\gamma - 1)}$$

where  $T$  is the absolute temperature and  $\gamma = 5/3$ . Find  $T_2/T_1$  that results in 30% efficiency ( $\eta = 0.3$ ).

#### 4.2. Secant Method



The natural frequencies of a uniform cantilever beam are related to the roots  $\beta_i$  of the frequency equation  $f(\beta) = \cosh \beta \cos \beta + 1 = 0$ , where

$$\beta_i^4 = (2\pi f_i)^2 \frac{mL^3}{EI}$$

$f_i$  =  $i$ th natural frequency (cps)

$m$  = mass of the beam

$L$  = length of the beam

$E$  = modulus of elasticity

$I$  = moment of inertia of the cross section

Determine the lowest two frequencies of a steel beam 0.9 m long, with a rectangular cross section 25 mm wide and 2.5 mm in. high. The mass density of steel is 7850 kg/m<sup>3</sup> and  $E = 200$  GPa.